Installation Manual

Trimble® Earthworks Grader

For SITECH Only

Version 2.24.x
Revision A
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Contents

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Tools and other supplies

1.1 Replacement serial number plates

If you need to install a system component over a serial number plate on the machine, you can order a replacement serial number plate from your dealer. You can then install the replacement serial number plate in another location on the machine. To maintain your machine's identification, you must obtain your dealer's approval to relocate the serial number plate before removing the existing serial number plate.

1.2 Tools

Make sure that you have the following tools available:

This item …	For…
Multimeter	Checking: • Power • Impedance across the CAN bus
Angle grinder	
Power drill	
Drill sets	
Battery chargers for all portable tools	
Spare battery packs for all portable tools – charged	
Extension cord	
Crimp tool and electrical terminals	Connecting the system to machine power
Deutsch crimper (P/N 5401-1568), Deutsch pins and sockets	Pinning wires for the Deutsch connectors
Diagonal cutters	
Wire strippers and pliers	
A length of pipe or wire	Feeding cables through the machine

This item ...	For...
Wrenches: • SAE • Metric open • Metric and imperial allen keys • Socket • Torque drivers	For fitting to specification
Steel tape or ruler	
Straight edge	
Framing square	
Digital level - 1.2 m (4 ft)	**NOTE –**A digital level is essential for a UTS system measure-up **NOTE –**Make sure the level is calibrated, in good working order and physical condition, and of good quality with a documented accuracy of 0.1°
Plumb line and string	
Hole saws	
Hacksaw	
Clamps	
Step ladder	Reaching the higher parts of the machine

1.3 Consumables

This item ...	For...
Assorted grinding discs	
Crimp terminals	Connecting the system to machine power
Deutsch pins and sockets	Pinning wires for the Deutsch connecters
Cable ties and wrap	Securing cables and preventing abrasion

This item ...	For...
Insulation tape	
Industrial strength hook and loop tape	
Red Loctite (P/N 33803) Blue Loctite (P/N 3207-1340)	Preventing screws and bolts from loosening
Anti-seize compound	
Electrical contact cleaner	
Glass cleaner	
Cleaning rags	
Rust prevention touch-up paint	Protecting welds and any machine parts stripped of paint during the install
RTV silicone sealer	Sealing cable entry holes
Small plastic bags	Protecting connectors from grease when feeding cables

2

Cab and body mount installation

In this chapter:

2.1 Overview of the install

This chapter provides best practices for installing sensors and GNSS antenna on the cab and body for motor graders for mastless configurations.

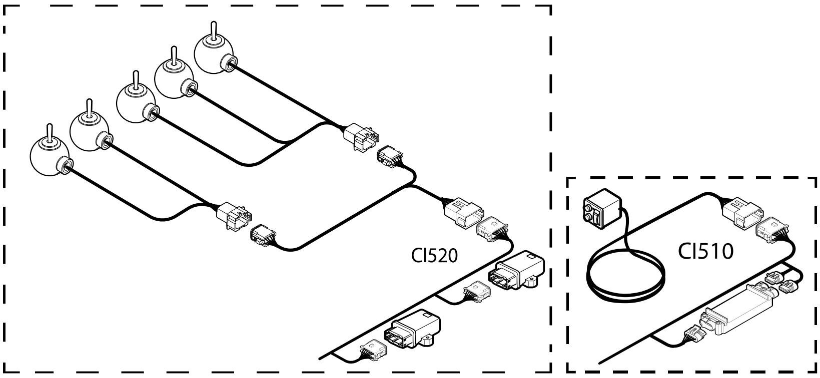
Use these best practices in conjunction with 120771 Trimble® AM CAT GRADE Mastless Upgrade for Motor Grader M/M2/M3/14A available on Partners . Before starting a cab and body install, review this entire chapter. This will ensure that you understand the best practices for a successful installation.

2.2 Cable routing best practice

When selecting a route for cables and harnessing consider the following:

- Exposure to hazards, including:
  - mechanical damage by the operator
  - environmental hazards
  - heat
  - corrosive material
- Moving parts
- Sharp edges

2.3 Cab components, cables and connections



2.4 IMU installation

Mount the circle shift sensor (yoke mounted IMU) with the connector facing down to reduce potential for water ingress.

2.5 EC520 installation

Align the EC520 parallel to the centerline of the goose neck. You can offset it to the left or right as long as it remains parallel.

M3 graders - mount the EC520 at the goose neck front. This avoids welding on critical parts of the goose neck and avoids interference with a mid-mount scarifier.

2.6 GNSS antenna installation

- Place the cab mounted GNSS antenna at least 500mm away from all transmitting antennas.
- Maximize the distance between the cab mounted GNSS antenna and the Cat product link antenna cluster to reduce iridium-band interference. On M3 machines use the existing bolt holes for mounting the GNSS antenna at the cab center rear and the SNR radio at the cab front.

2.7 Link bar pin selection

For best accuracy and side shift extension, use the center hole for the link bar pin. If the link bar pin is changed, a new measure-up is required.

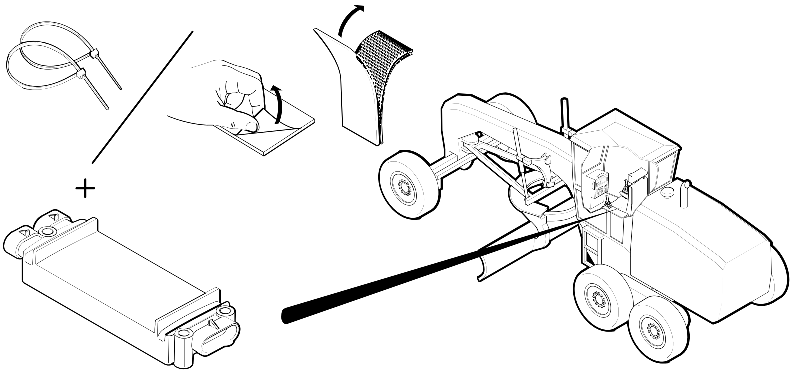
2.8 CI5xx CAN interface module install

The CI5xx is a Controller Area Network (CAN) adapter that that converts the remote switch increment and decrement, and sideshift inputs to CAN.

Both the CI510 and the CI520 CAN interfaces are supported. The CI510 supports up to 8 switches. The CI520 supports up to 4 switches. Use two CI520s to connect 5-8 switches.

- Attach CI510
- Attach CI520

2.8.1 Attach CI510



2.8.2 CI510 connections, components, and cables

Part number	Items	Description
150906-01	1	Valve module harness
86930-05	1	Valve cable
09231-10	1	CI510 CAN Interface Module
150754-01		Switch harness

2.8.3 Install the CI510 CAN interface module

Have the following parts available:

- Dual locking strip
- Cable ties

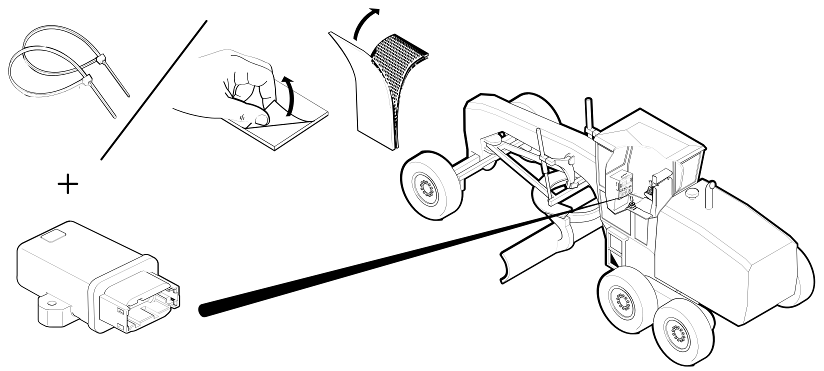
To install:

1. Connect the CI510 to the cabling as shown in the connection diagram. See 2.3 Cab components, cables and connections.
2. Place the CI510 inside, or under, the cab. It is often behind the seat.
3. Either use zip ties or use the Velcro supplied in the kit to secure the CI510, if required.

2.8.4 Validate

After connecting the display and running Earthworks, select System Status > Remote Switches. The remote switch status should be Active and the model should be CI510.

2.8.5 Attach CI520



2.8.6 CI520 connections, components, and cables

Part number	Items	Description
150906-01	1	Valve module harness
86930-05	1	Valve cable
132739-10	1	CI520 CAN Interface Module
150820-01 Rev B		Switch harness

2.8.7 Install the CI520 CAN interface module

Have the following parts available:

- Dual locking strip
- Cable ties

To install:

1. Connect the CI520 to the cabling as shown in the connection diagram. See 2.3 Cab components, cables and connections.

   > **NOTE –**The CI520 has only four switch inputs. Use two CI520s when up to eight switch inputs are needed.

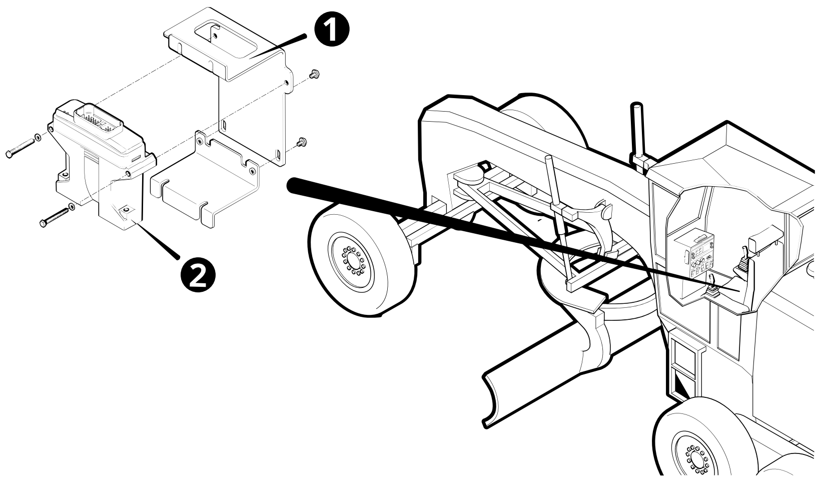
2. Place the CI520 inside, or under, the cab. It is often behind the seat.
3. Either use zip ties or use the Velcro supplied in the kit to secure the CI520, if required.

2.8.8 Validate

After connecting the display and running Earthworks, select System Status > Remote Switches. The remote switch status should be Active and the model should be CI520.

2.9 Valve module install

The valve module provides precise control of the blade's lift and tilt functions by sending output signals to the machine's electro-hydraulic valves or the hydraulic kit installed on the machine as part of this system.



Item	Part number	Description
1	196540-10	Module mounting bracket
2	100441-00	VM510

2.9.1 Connections, components, and cables

Part number	Items	Description
150906-01	1	Valve module harness
86930-05	1	Valve cable
150858-01	2	Splitter, 4 Circuit
150473-020	1	Cable extension

2.9.2 Install the valve module

1. Attach the valve module to the module mounting bracket.
2. Connect the cabling to and from the valve module according to the connection diagram for generic dozers, available on Partners.

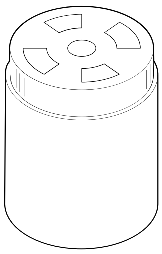
2.9.3 Validate

After connecting the display and running Earthworks, use the Web Interface's Configure > Valve Test screen to test that the valves are driving the lift and tilt rams.

2.10 Installing the AA510 audible alarm

The AA510 alerts the operator to whether they are on grade, above grade or below grade. The AA510 attaches to the system via a CAN bus.

2.10.1 Attach AA510



- Rubberized p-clamp P/N 97587-45
- Cable ties

To install:

1. Decide where to locate the AA510.
2. Connect the AA510 to the EC520 long platform harness, as shown in the connection diagram.
3. Use P-clamp or cable ties to attach the AA510 to a fixed surface or component.

2.10.2 Validate

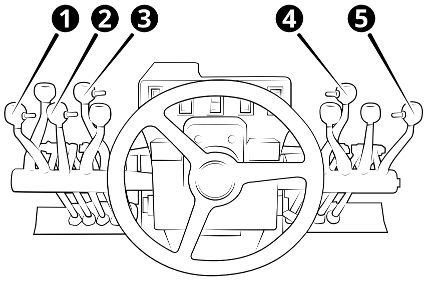
After connecting the display and running Earthworks, test the AA510 using the System Settings > Sound option in Earthworks.

Some of the problems that you might see include:

Problem	Description	Resolution
AA510 too quiet or too loud.	The AA510 might seem loud in one machine and not loud enough in another machine because of where you locate the AA510 and the machine noise.	In the Operator App, open Sound Settings and edit the volume of the AA510 and the type of event that sets off the AA510.

2.11 Remote switch install

The motor grader system can have up to five remote switches:



1	Left Auto switch on left blade lift lever.	2	Sideshift Auto switch on sideshift lever.
3	Elevation offset / target cross slope switch on moldboard roll lever.	4	Elevation offset / target cross slope switch on articulation lever.
5	Right Auto switch on right blade lift lever.		

2.11.1 Install the switch harness

Select a route for the cable by considering:

- Exposure to hazards, including:
  - mechanical damage by the operator
  - environmental hazards
  - heat
  - corrosive material
- Access to the CI5xx cable
- Moving parts
- Sharp edges

2.11.2 Install and connect the remote switches

To install:

1. Have P/N 161851-99-30 available.
2. Remove the knobs from the levers as shown above.
3. Replace with the remote switches
4. Route and secure the remote control switch and connect the cable to the cab harness.

2.11.3 Validate

Some of the problems that you might see include:

Problem	Description	Resolution
Remote switch does not work.	The switch is not connected.	Check the switch connects system by opening the Web Interface and checking Monitor > Onboard Devices. Check the connections.

3

EM410 electric mast installation

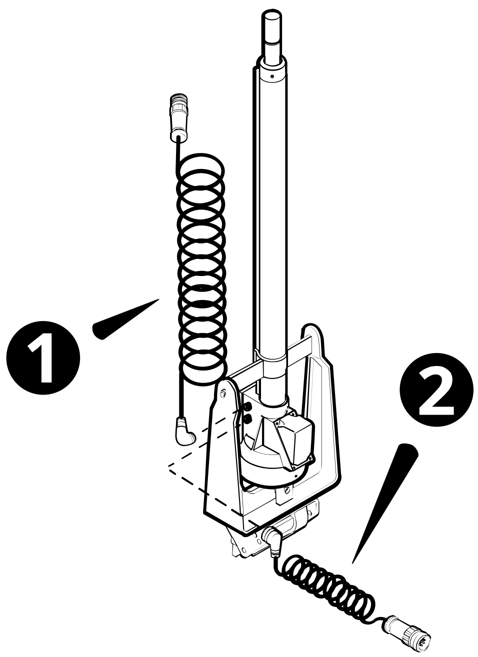
In this chapter:

3.1 Overview of the install

This chapter describes how to install an electric mast, an M4 shock mount, and an adjustable mast mount.

Before starting an electric mast install, review this entire chapter. This will ensure that you understand the full installation workflow. It also lets you consolidate welding and fitting tasks, which may require external contractors, with other subsystem installations, such as the cab and blade installations.

3.2 Electric mast connections, components, and cables



Item	Part number	Description
❶	0312-0990	Coil cable with right angle connector
❷	150441-030	Coil cable with quick disconnect connector *OR*
	150445-030	Coil cable with strain relief and quick disconnect connector

3.2.1 Components and cables from the mast mount base kit (P/N 0791-1740)

Part number	Items	Description
0791-1740	1	Adjustable mast mount with mounting bolts
2810-1468	4	Blanking screws

3.2.2 Components and cables from the M4 shock mount kit (P/N 0395-8270)

Part number	Items	Description
0395-3000	1	Shock mount with mounting bolts

3.2.3 Additional components and cables

Part number	Required	Description
109966-00	1 or 2	EM410 electric mast
0312-0990	1 or 2	Coil cable with right angle connector (❶)
150441-030 *OR*	1 or 2	Coil cable with quick release connector (❷)
150445-030		Coil cable with strain relief and quick release connector (❷)
0791-2220	1 or 2	Weld bracket

3.3 Install the adjustable mast mounts

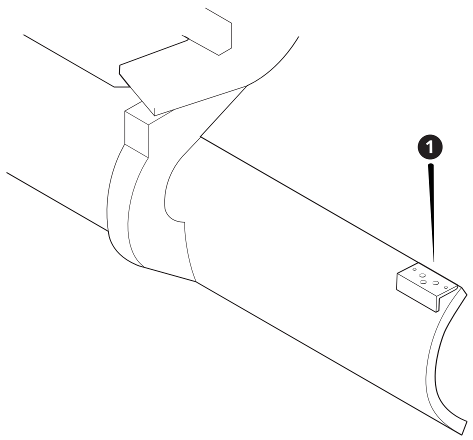
Adjustable mast mounts are used on machines where the blade is capable of a significant amount of roll or pitch.

3.3.1 Install the mast mount bracket

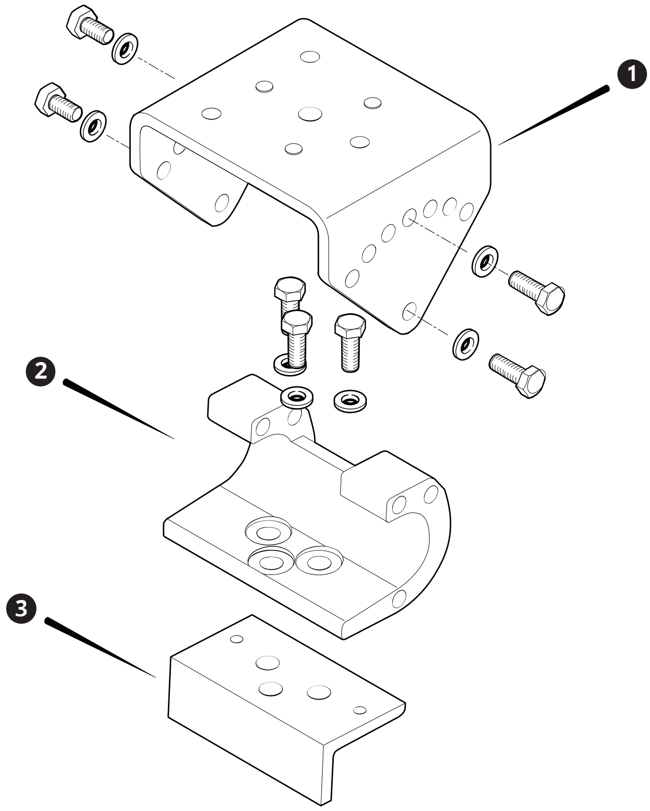
When you select a location for the adjustable mast mount brackets, consider the following:

- For two masts, install the brackets symmetrically on the blade. For one mast, install the bracket on the left end of the blade (when viewed from the cab).
- Position the brackets approximately the same distance in from the ends of the blade, and the same distance above the cutting edge bolts.
- Make sure that the brackets are square with the cutting edge.
- If you intend to use sonic tracers, take the sonic tracer pole clamp mounting positions into consideration.
- Fix one edge of the bracket to the top of the blade, and the other to the back of the blade.
- Make sure you allow enough slack in the cable so that the operator can swing the yoke and blade out to the pin position for slope banking applications.

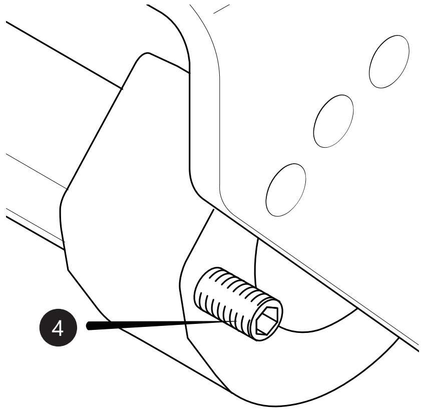
Tack weld the bracket (❶, P/N 0791-2220) to the blade:



3.3.2 Install the adjustable mast mount



1. Attach the lower section of the mast mount (❷) to the bracket (❸).
2. Torque the bolts to 370 ± 50 N·m (273 ± 37 lb-ft).
3. Attach the upper section (❶) to the lower. For accurate guidance, you ***must*** position the upper section so that the top of the section is level (parallel to the cutting edge).
4. Depending on the amount of roll or pitch expected for the blade, identify the bolt hole on the lower section that the bolt securing the mount will engage with. Note:
   - You can use any of the seven bolt holes in the upper section with either of the two bolt holes in the lower section.
   - You ***must*** use the supplied blanking screw (❹)to fill the unused hole on the lower section:



5. When you are satisfied that everything is correct, weld the bracket (❸) securely.

3.4 Install the electric masts

1. Adjust the adjustable mast mount until it is approximately level, and secure with a bolt through the hole in the top section of the adjustable mast mount that comes closest to overlapping the open hole in the lower section.

   Make a note of the chosen bolt hole in the upper section. During commissioning, the bolt holes in the upper section are identified with numbers from 1, nearest the front of the machine to 7, nearest to the cab.
2. Assemble the M4 shock mount following the instructions in the Parts Bulletin (P/N 0395-3060) included in the M4 shock mount kit.
3. Bolt the shock mount to the adjustable mast mount. The shock absorbing components should be parallel to the centerline of the machine.
4. Place the electric mast onto the base bracket of the shock mount, and secure the mast into the shock mount using the mast clamp.
5. Securely bolt the base of the mast to the base bracket of the shock mount.

4

ST400 sonic tracer mount installation

In this chapter:

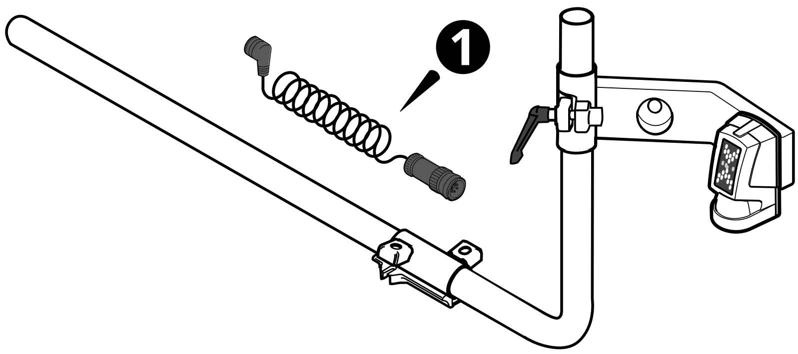
4.1 Overview of the install

This chapter describes how to install the components for a sonic tracer mount onto the blade.

Before starting a sonic tracer install, review this entire chapter. This will ensure that you understand the full installation workflow. It also lets you consolidate welding and fitting tasks, which may require external contractors, with other subsystem installations, such as the cab and blade installations.

> **TIP –** For some simple, small laser applications, you can mount an LR410 on the sonic pole mount. For best results on more complex, or larger laser applications, the recommended mounting for an LR410 is an electric mast.

4.2 Sonic tracer connections, components, and cables



Item	Part number	Description
1	0794-2350	Coil cable with quick disconnect connector

4.2.1 Components and cables from the sonic tracer mounting kit (P/N 0395-9870)

Part number	Items	Description
0395-0880	1	Sonic tracer add-on kit. This kit includes: • Coil cable with quick disconnect connector (❶) • Sonic tracer mounting arm • Pole clamp components and assembly instructions
66481-10	1	Pole

4.2.2 Additional components and cables

Part number	Required	Description
59701-90 *OR* 59701-91	1 *OR* 2	ST400 sonic tracer

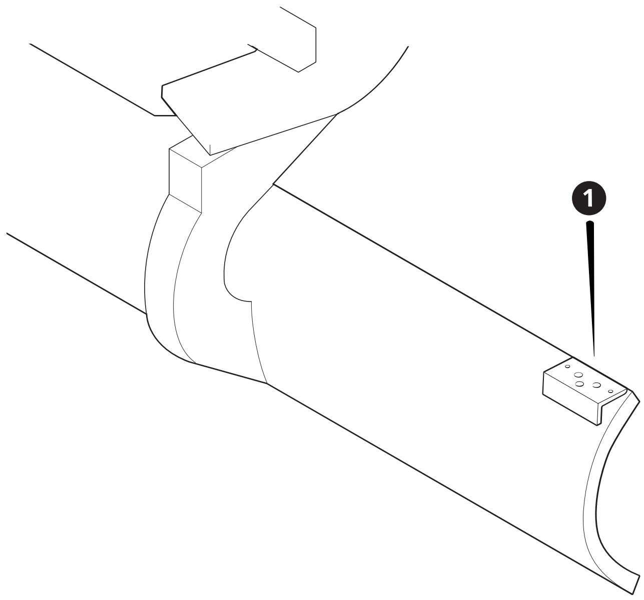
4.3 Clamp bracket

A sonic tracer pole clamp bolts on to the same type of mounting plate as that used by an adjustable mast mount.

When you select a location for the pole clamp brackets, consider the following:

- If you intend to use electric masts, take the electric mast mounting positions into consideration.
- The sonic tracer mounting tube extends past the end of the cutting edge. This is to ensure that the sonic tracer sits vertically above the reference surface or stringline.
- For graders, ensure the sonic tracer mounting tube clears the circle when you sideshift the moldboard.
- Fix one edge of the bracket to the top of the blade, and the other to the back of the blade.
- Make sure you allow enough slack in the cable so that the operator can swing the yoke and blade out to the pin position for slope banking applications.

Weld the bracket (❶, P/N 0791-2220) to the blade: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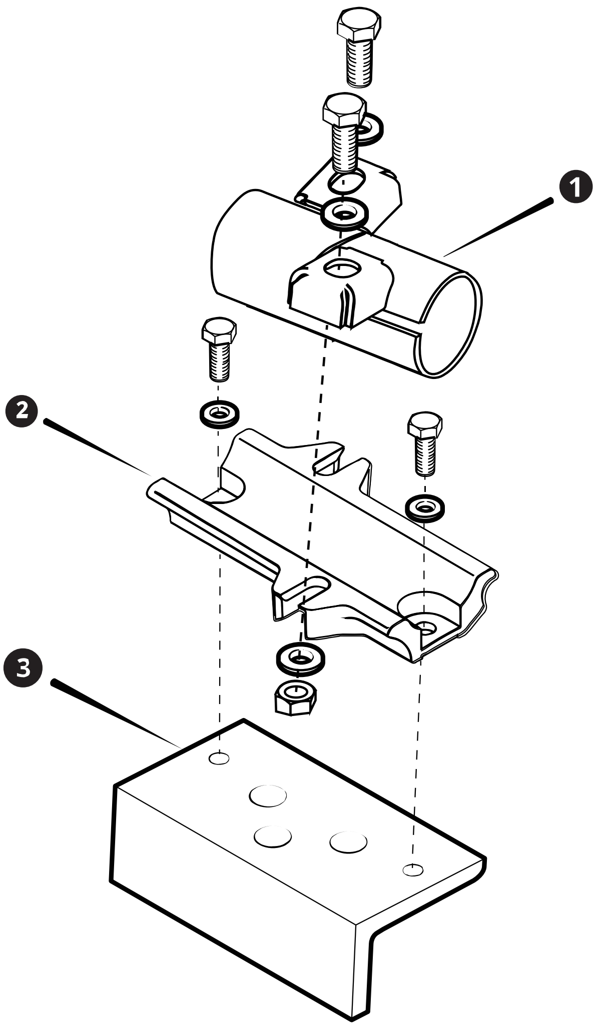
4.3.1 Sonic tracer pole clamp



1. Bolt the clamp base (❷) to the bracket on the blade (❸).
2. Loosely bolt the pole clamp (❶) to the clamp base (❷).
3. Insert the long arm of the pole into the clamp, and with the short arm roughly vertical, tighten the pole clamp bolts.

5

Torque charts

5.1 Imperial

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

			Nominal size (inches)								
	Grade	Torque	1/4	5/16	3/8	7/16	1/2	9/16	5/8	3/4	1”
	SAE 2	FT-LB	6	11	19	30	45	66	93	150	300
		Nm	8	15	26	41	61	89	126	203	407
	SAE 5	FT-LB	9	18	31	50	75	110	150	250	580
		Nm	12	24	42	68	102	149	203	339	786
	SAE 8	FT-LB	13	29	46	75	115	165	225	370	890
		Nm	18	39	62	102	156	224	305	502	1207
	SOCKET HEAD	FT-LB	14	30	50	80	120	175	240	390	960
		Nm	19	41	68	108	163	237	325	529	1302

5.2 Metric

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

			Nominal size (mm)							
	Grade	Torque	6	8	10	12	14	16	18	20
	4.6, 4.8	FT-LB	3	8	16	27	43	60	90	120
		Nm	4	11	22	37	58	81	122	163
	8.8	FT-LB	7	17	33	59	100	130	180	280
		Nm	10	23	45	80	136	176	244	380

			Nominal size (mm)							
	Grade	Torque	6	8	10	12	14	16	18	20
10.9	10.9	FT-LB	10	25	48	85	130	200	280	400
		Nm	14	34	65	115	176	271	380	542
12.9	12.9	FT-LB	12	29	57	100	150	230	–	–
		Nm	16	39	77	136	203	312	–	–

Notices

See https://heavyindustry.trimble.com/earthworksreleases for the latest:

- Terms and conditions.
- Important safety information.
- Warranty and compliance information.

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